

Name: Carson, Max, Olivia, Sophia

Date: 02/09

Class: Gold Day 8

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1

- For our highest priority, it is the size. Our size is 16 by 16 inches. The size is very important because this allows us to know how many things we are allowed to implement into our design.
- Because we need to know what we are allowed to implement into our design.

Design Requirement #2

- The second highest priority for the design requirement is acquiring a fan. This is essential for taking in air and stale air out.
- This design requirement ties directly to the prior research because it is essential to have an airflow intake and eject air.
- This fan must be battery-powered (rechargeable batteries are an option) or plugged in. This fan needs an electric current like that due to the air purifier being static in one place, and not moving like a vehicle air intake that runs due to the motion of the object.

Design Requirement #3

- Our 3rd Highest priority for our design is to have it chargeable.
- So at the end of the day, the customer can grab the batteries and put them on the charger so they are charged for the next day.

Design Requirement #4

- We wanted our design to be long lasting because even though our intent is for it to be easy to change/manage we wanted it to not have to be changed every week. We thought of this because we thought if you had to change it every week it would take time out of the class time.
- A long-lasting air filter can last for about 6 months before you have to replace it. Cheap air filters last for about 30 days. Long-lasting filters are normally pleated filters. With pleated filters, you won't need to replace it every month.

Design Requirement #5

- The 5th most important priority is ensuring that it is easy to manage. Changing materials such as the filters or batteries should not be difficult at all.

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- This design ties to our research because our stakeholders have asked for the filter to be easy to access and change so anyone can do it if they are busy.

Design Requirement #6

- Our 6th LOW Priority is for our filter to be quiet and not cause distractions while in use in the classroom.
- We don't want our filter to be a distraction to the teacher and students as they are learning.

Design Requirement #7

- A way to clear the air and make the scent of the room or area better is by using a diffuser that releases essential oils.
- This helps with our design because after the air is cleared, rooms smell dull. So diffusers make the air smell better. Online, websites say that it will moisturize the room since after using an air filter can be sometimes dry. Many past students also mention how the room smells bad after working on their projects.